Cross-fitted selective recovery under controlled multi-sensor corruption

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Abstract

An available sensor group can remain in the inference path after its representation has been corrupted. A recovery model may then correct a wrong decision or overturn a correct one. We address this trade-off with paired clean–faulted training and a cross-fitted router that selects between a bounded base model and a recovery-supervised endpoint without test labels or fault-type inputs. The primary deployment objective is hard-decision safety, measured by negative-transfer prevention, recovery retention and net utility; ranking and probability quality are secondary outcomes. On 1,436 held-out observations per mechanism with a controlled fault that reached an available group, the two-pass Lite-CF router prevented 94.7% of recovery-induced negative transfers while retaining 9.7% of available corrections. Its mean macro-AUROC was 0.8151, compared with 0.8115 for the base model and 0.8175 for unconditional recovery. Router calibration was sparse and model-specific: only 89.7 correctness disagreements informed each fit on average, and repeated thresholds ranged from 0.525 to 0.90. Mechanism-fixed intervals supported a positive Lite-CF effect for Gaussian and stuck-at corruption, but not uniformly for offset or drift. At equal correction and harm values, Lite-CF gained 15.8 correct decisions per 10,000 strict observations; this descriptive advantage disappeared when one harm exceeded approximately 2.9 corrections. Unsupported deployments must retain the base endpoint. The evidence supports a conservative risk-control layer for controlled available-group corruption, not general predictive superiority, natural-failure recovery or device-independent field performance.

Keywords: multi-sensor fusion; sensor corruption; fault-tolerant classification; selective recovery; negative transfer; cross-fitting

1 Introduction

Scientific and industrial systems increasingly combine several channels whose availability and fidelity can change after deployment. Missing inputs, delayed diagnostics and distribution shifts can invalidate a model even when its architecture performs well on a static benchmark [1–4]. This problem is distinct from ordinary random missingness because a faulty sensor can remain present and produce plausible but biased, clipped, noisy or unresponsive measurements. A robust fusion rule must therefore decide not only whether a channel exists, but how strongly it should influence the prediction.

Several training strategies improve tolerance to absent or inconsistent sources. Modality dropout, masking and self-distillation expose models to missing inputs [5–7]. Representation refinement, gradient decoupling and expert routing address source imbalance or train–test mismatch [8–11]. Reliability-aware fusion additionally uses predictive uncertainty, learned credibility or rejection [12–17]. Quality-aware multimodal fusion (QMF) provides a published failure-aware comparator by deriving modality confidence from predictive energy and aligning confidence with historical loss trajectories [18]. These approaches establish useful design patterns, but they do not make an arbitrary weighting response physically calibrated or comparable across sensors.

Sensor-fault diagnosis has a complementary engineering literature. Redundant sensor networks can localize offset, drift, noise, peaks and stuck responses, and can distinguish local sensor faults from global system changes [19–21]. Hydraulic condition-monitoring studies further show the value of heterogeneous pressure, flow, temperature, power and vibration measurements for component-state classification [22–25]. These studies motivate physically defined sensor groups and independent condition-monitoring validation rather than arbitrary computational groupings alone.

Most open benchmarks nevertheless encode sensor failure through simulation, laboratory intervention or post hoc feature corruption. A recent public AHU resource instead provides one year of building-management-system records from an office, an auditorium and a hospital, with return- and supply-air temperature sensor faults annotated by experienced HVAC engineers [26, 27]. These labels create a useful external field test, although they are rule- and expert-adjudicated operational annotations rather than maintenance-confirmed hardware replacement events.

Recent AHU studies show that the direct diagnosis problem differs from fault-tolerant recovery. Transformer models can exploit long temporal dependencies within a continuously operating hospital, while attention-based transfer and temporal–contextual learning explicitly address building shift and limited labels [28–30]. These architectures preserve fault signatures to identify the failure itself. In contrast, fault-tolerant classification seeks to suppress corrupted evidence while retaining another target, so superiority in one task need not transfer to the other.

We study paired-degradation responsive fusion (PDRF). "Responsive" denotes an operational change in fusion weight after a controlled sensor degradation; it does not imply physical health, calibrated risk or sensor variance. Each training observation is paired with a view in which one available sensor group is degraded. The model learns a bounded response, and directional losses encourage the affected sensor's response to rise and its fusion weight to fall. The formulation does not invoke an identifiable causal counterfactual. Predictive calibration is evaluated separately because discrimination, uncertainty and calibration are different properties [31–37].

The study makes four contributions. First, it defines a strict recovery estimand that includes only controlled interventions that reach an available sensor group. Second, it formulates selective recovery as hard-decision risk control and cross-fits both the routing score and operating threshold. The primary Lite-CF implementation uses six endpoint features and two forward passes. Third, it audits the sparse routing problem through repeated grouped splits, coefficient and sign stability, regularization paths, calibration-size sensitivity, an all-row three-outcome model, a shallow tree and a staged cascade. Fourth, it separates mechanism-fixed, batch-fixed and fitting-seed variation and reports the final routed probabilities on discrimination and calibration metrics. The title and claims are limited to controlled corruption. Hydraulic results are single-rig sensitivity evidence, and expert-annotated air-handling-unit diagnosis remains a negative task boundary because it lacks an independent downstream recovery target.

2 Related work

2.1 Missing and unreliable sensor inputs

Missing-data mechanisms affect the interpretation of any robustness experiment [38]. Random sensor dropout tests tolerance to absent channels, whereas the present study targets channels that remain available but become corrupted. Corruption augmentation and consistency distillation can improve this setting, but they can also make an apparently novel fusion rule indistinguishable from robust training. Our comparator design therefore gives the main gates identical paired degradations, labels, minibatches and additional forward passes, and separately evaluates early and uniform fusion with clean distillation and modality-dropout self-distillation.

2.2 Bounded operational responses and probability calibration

Input-dependent losses can represent aleatoric variation under valid probabilistic assumptions, whereas PDRF uses its bounded response only for weighting [31, 32]. Response ordering and isotonic audits therefore do not convert it into physical variance. Probability quality is assessed separately with NLL, Brier score, multiple ECE definitions and temperature scaling because calibration depends on both multiclass structure and binning [37, 39, 40]. Entropy weighting, deep ensembles and conformal sets provide uncertainty-oriented comparators, not device-health estimates [33, 41]. Cross-building AHU controls likewise distinguish source-only GroupDRO from target-feature adaptation by CORAL or target normalization [42–44].

2.3 Selective prediction and model fallback

Selective classification and reject-option learning trade coverage for lower predictive risk [45–47]; recent work additionally studies selection under distribution shift [48]. Learning-to-defer methods route difficult cases to a human or another expert whose error pattern may differ from that of the primary predictor [49, 50]. Classifier cascades explicitly trade predictive accuracy against evaluation cost, while dynamic classifier and ensemble selection estimate which member is locally competent [51, 52]. Mixture-of-experts routing provides a related differentiable allocation mechanism [14]. The present fallback problem is narrower: both endpoints are fixed models of the same sensor-fusion task, no observation is rejected, and the selector specifically controls recovery-induced negative transfer while retaining corrections. Its operating curve is therefore prevention versus retained recovery, rather than conventional risk versus coverage, human deferral utility or average cascade cost.

2.4 Evidence and statistical estimands

Classifier comparisons can change with the resampling unit and evaluation estimand [53, 54]. Optimization seeds describe repeated fitting, whereas sample bootstrap intervals describe a fixed set of fitted predictors [55]. McNemar's paired test evaluates discordant hard decisions on the same observations [56]. We report these levels separately and do not treat observations from one instrument as independent devices. Precision–recall metrics are included because class imbalance can make ROC summaries incomplete [57].

3 Methods

3.1 Problem definition

For observation i and sensor group m, let $\mathbf{x}_{im} \in \mathbb{R}^d$ be the feature vector, $a_{im} \in \{0,1\}$ the availability indicator, $q_{im} \in (0,1]$ an optional diagnostic-quality covariate and $y_i \in \{1,\ldots,C\}$ the target. PDRF estimates

$$p\big(y_i \mid \{\mathbf{x}_{im}, a_{im}, q_{im}\}_{m=1}^{M}\big)\,. \tag{1}$$

When no quality metadata are available, all q_{im} equal one. Every evaluated observation contains at least one available group. If all groups are unavailable, the implementation abstains before inference because normalized fusion is undefined.

For the routing audit, let b_i indicate that the base endpoint is correct, r_i that the recovery endpoint is correct and s_i that the router emits the recovery output. Table 1 fixes the event terminology used throughout.

Table 1: Hard-decision event definitions. "Prevention" is one minus the fraction of harm opportunities that are realized. A harmful switch and a residual harm are the same event expressed per observation rather than per harm opportunity.

Quantity	Event definition	Reported rate
Correction	$\neg b_i \wedge r_i$	retained corrections / corrections
Negative transfer	$b_i \wedge \neg r_i$	prevented harms / harm opportunities
Residual harm	$b_i \wedge \neg r_i \wedge s_i$	residual harms / observations
Recovery selection	s_i	selected recovery outputs / observations
Harmful switch	$b_i \wedge \neg r_i \wedge s_i$	harmful switches / observations
Net correct change	corrections retained − harms realized	events per 10,000

3.2 Architecture and bounded weighting

Each sensor group is encoded by affine layers $d \to 64 \to 32$, with a ReLU after each hidden affine transformation and no normalization or dropout. A biased linear $32 \to C$ output produces group-specific logits $\boldsymbol{\mu}_{im}$. The response head is exactly $\mathrm{Linear}(32, 16, \mathrm{bias} = \mathrm{true})$, ReLU, then $\mathrm{Linear}(16, 1, \mathrm{bias} = \mathrm{true})$. There is no activation after the final linear layer, so its raw response u_{im} can be positive or negative. The bounded operational response is

$$\begin{aligned} r_{im} &= B \tanh(u_{im}/B), \\ -B &< r_{im} < B. \end{aligned} \tag{2}$$

where the prespecified main setting uses $B = 3$. This layer placement matches the released implementation and gives the symmetric open interval in Eq. (2). Across 17,650 clean and faulted seed–observation responses from 10 fitted RO-PDRF-Full models, the observed float32 range was −2.899454 to 3.000000; the latter is numerical rounding of a value below the mathematical open boundary. Clean responses ranged from −2.757962 to 2.997449. Unnormalized and normalized weights are

$$\begin{aligned} \widetilde{w}_{im} &= a_{im} q_{im} \exp(-\beta r_{im}), \\ w_{im} &= \frac{\widetilde{w}_{im}}{\sum_k \widetilde{w}_{ik}}. \end{aligned} \tag{3}$$

with $\beta = 0.25$. Fused logits are $\boldsymbol{\ell}_i = \sum_m w_{im} \boldsymbol{\mu}_{im}$. The implementation applies a 10^{-6} lower guard to the normalization denominator, but the guard is inactive under the stated input constraints because every observation has at least one available group and $q_{im} \geq 0.01$ after clipping.

This construction differs from an unconstrained softmax gate in three testable ways. First, the weight is factorized into availability, an optional externally measured quality term and one bounded operational response; it is not an unrestricted function of the concatenated sensor representation. Second, the perturbed group is known during paired degradation, so its response and normalized weight receive directional supervision rather than only a task-gradient signal. Third, the same response that controls fusion is regularized by a group-specific predictive loss. Thus the response is operationally coupled to group evidence, although it is not asserted to be a physical health variable.

The factorization also gives two elementary stability properties. For an available group, fixed q and any $k \neq m$,

$$\begin{aligned} \frac{\partial w_{im}}{\partial r_{im}} &= -\beta w_{im}(1 - w_{im}) \leq 0, \\ \frac{\partial w_{ik}}{\partial r_{im}} &= \beta w_{ik} w_{im} \geq 0. \end{aligned} \tag{4}$$

For two available groups with equal quality, bounded responses imply

$$e^{-2\beta B} < \frac{\widetilde{w}_{im}}{\widetilde{w}_{ik}} < e^{2\beta B}. \tag{5}$$

At $B = 3$ and $\beta = 0.25$, one response alone can therefore create at most a 4.48-fold unnormalized-weight ratio between equal-quality groups. These properties do not guarantee that the learned ordering is correct; the experiments separately test that question.

The response regularizer, retained as $\mathcal{L}_{\text{score}}$ in the implementation for reproducibility, combines a bounded heteroscedastic-style term and a boundary penalty. For a minibatch $\mathcal{B}$, write $s_{im} = \omega_{y_i}[-\log \operatorname{softmax}(\boldsymbol{\mu}_{im})_{y_i}]$, where $\omega_c = n/(Cn_c)$ is the training-set class weight, and let $h_{im} = e^{-r_{im}} s_{im} + r_{im}$. Then

$$\mathcal{L}_{\text{score}} = \frac{\sum_{i \in \mathcal{B}} \sum_m a_{im} h_{im}}{\sum_{i \in \mathcal{B}} \sum_m a_{im}}, \tag{6}$$

$$\mathcal{L}_{\text{bound}} = \frac{\sum_{i \in \mathcal{B}} \sum_m a_{im} (r_{im}/B)^4}{\sum_{i \in \mathcal{B}} \sum_m a_{im}}. \tag{7}$$

Because cross-entropy is non-negative and $r_{im} > -B$, $\mathcal{L}_{\text{score}} > -B$. The complete objective therefore has a finite lower bound. This statement concerns optimization stability, not calibration of the bounded operational response.

3.3 Paired degradation and directional losses

For each minibatch observation, one available group m^* is drawn from a fixed training prior. A retention factor $\rho_i \sim \text{Uniform}(0.15, 0.50)$ controls additive Gaussian degradation with scale $2.5(1 - \rho_i)$. In half of the pairs, q_{im^*} is multiplied by ρ_i; otherwise it is unchanged. Let primed symbols denote the degraded view and $d_i = 1 - \rho_i$.

The clean and degraded task losses use the class-weighted mean implemented by PyTorch. With $e_i(\boldsymbol{\ell}) = -\log \operatorname{softmax}(\boldsymbol{\ell})_{y_i}$,

$$\begin{aligned} \mathcal{L}_{\text{task}} &= \frac{\sum_{i \in \mathcal{B}} \omega_{y_i} e_i(\boldsymbol{\ell}_i)}{\sum_{i \in \mathcal{B}} \omega_{y_i}}, \\ \mathcal{L}_{\text{deg}} &= \frac{\sum_{i \in \mathcal{B}} \omega_{y_i} e_i(\boldsymbol{\ell}'_i)}{\sum_{i \in \mathcal{B}} \omega_{y_i}}. \end{aligned} \tag{8}$$

Let $\mathcal{V}_{\mathcal{B}} = \{i \in \mathcal{B} : \sum_m a_{im} > 1\}$ denote observations for which the selected group can be removed while another group remains. For $i \in \mathcal{V}_{\mathcal{B}}$, the removal prediction $\mathbf{p}_i^{(-m^*)}$ defines a severity-interpolated target $\mathbf{t}_i$; write $\widehat{\mathbf{t}}_i = \text{stopgrad}(\mathbf{t}_i)$:

$$\begin{aligned} \mathbf{t}_i &= (1 - d_i)\mathbf{p}_i + d_i \mathbf{p}_i^{(-m^*)}, \\ \mathcal{L}_{\text{cons}} &= \frac{1}{|\mathcal{V}_{\mathcal{B}}|} \sum_{i \in \mathcal{V}_{\mathcal{B}}} \|\mathbf{p}'_i - \widehat{\mathbf{t}}_i\|_2^2. \end{aligned} \tag{9}$$

If $\mathcal{V}_\mathcal{B}$ is empty, $\mathcal{L}_{\text{cons}}$ and the removal-dependent directional terms are omitted, equivalently set to zero. Directional terms operate on the affected normalized weight and raw response and are averaged over eligible observations. With $[z]_+ = \max(0, z)$ and $\Delta w_i = w'_{im^*} - w_{im^*}$,

$$\mathcal{L}_{\text{mono}} = |\mathcal{V}_\mathcal{B}|^{-1} \sum_{i \in \mathcal{V}_\mathcal{B}} [\Delta w_i]_+. \tag{10}$$

With $\gamma_i = 0.5 d_i / 0.85$ and $\Delta r_i = \text{stopgrad}(r_{im^*}) - r'_{im^*}$, the response-ranking term is

$$\mathcal{L}_{\text{rank}} = |\mathcal{V}_\mathcal{B}|^{-1} \sum_{i \in \mathcal{V}_\mathcal{B}} [\gamma_i + \Delta r_i]_+. \tag{11}$$

The final loss is

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\text{task}} &+ 0.20\mathcal{L}_{\text{score}} \\ &+ 0.50\mathcal{L}_{\text{bound}} + 0.15\mathcal{L}_{\text{deg}} \\ &+ 0.10\mathcal{L}_{\text{cons}} + 0.05\mathcal{L}_{\text{mono}} \\ &+ 0.15\mathcal{L}_{\text{rank}}. \end{aligned} \tag{12}$$

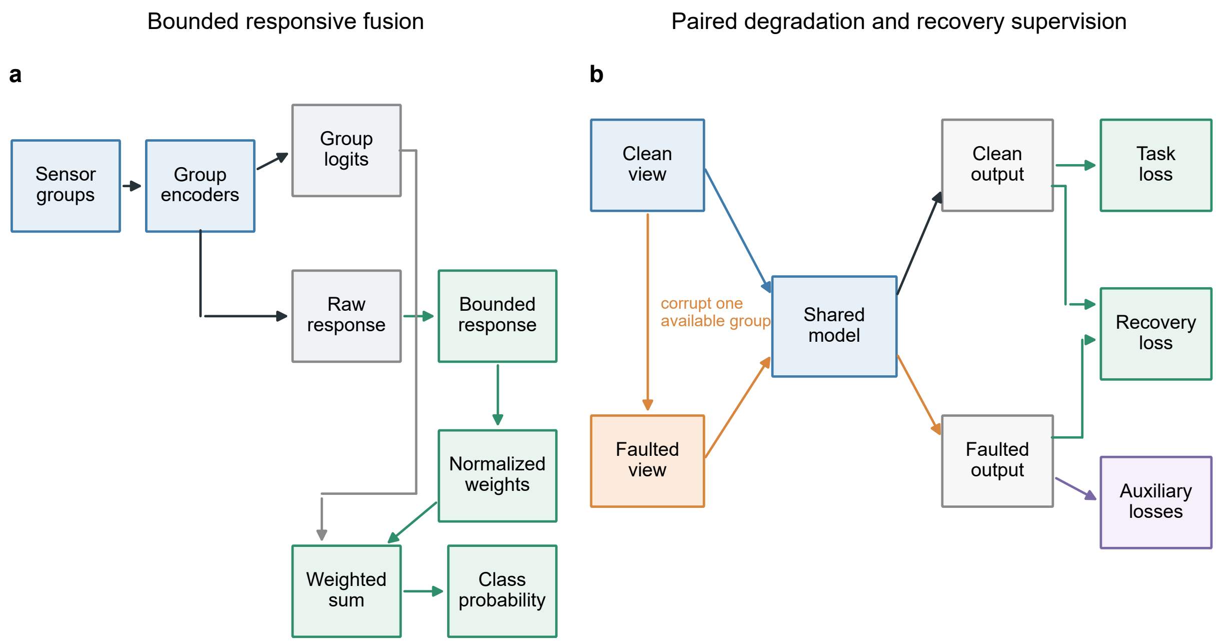

Figure 1: **Bounded responsive fusion and recovery supervision.** **a**, Each sensor group produces class logits and a bounded operational response. Availability, optional diagnostic quality and the response determine normalized fusion weights. Distinct connector ports separate the logit and weight contributions to the fused prediction. **b**, Clean and faulted views share model parameters. The detached clean prediction supervises recovery of the faulted prediction; proper-scoring and ranking terms are auxiliary controls.

3.4 Comparators and mechanism ablations

Early fusion concatenates zero-filled group features, masks and quality values before a $128 \to 64 \to C$ multilayer perceptron. Uniform fusion averages group logits. EF-PD and UF-PD receive the same paired degradations and degraded-task loss as PDRF. The corruption-aware gated fusion model (CAGF) concatenates the four 32-dimensional group representations, masks and quality values, then applies a $64 \to M$ softmax gate. It is the matched mixture-of-experts-style comparator: group classifiers act as experts and a sample-dependent unrestricted gate supplies their mixing coefficients. CAGF receives the same paired degradation, consistency and normalized-weight monotonicity losses. Direct weight ranking (DWR) adds a margin directly to the gate weight. The quality-only model uses $a_{im}q_{im}$ without a learned gate or bounded operational response.

To test whether a stronger interaction model alters the architecture comparison, RO-AT-GATE maps each physical sensor group to a 32-dimensional token, adds a learned group-identity embedding, and applies one four-head self-attention layer with a 64-unit feed-forward block. Unavailable groups are masked before attention. A shared $32 + 2 \to 16 \to 1$ head converts each contextualized token, availability flag and quality value into an unrestricted softmax gate. This compact attention baseline has 26,809 parameters, close to CAGF's 26,588, and receives exactly the same paired views and complete recovery objective as RO-CAGF. It is a sensor-group adaptation of modern cross-modal attention practice rather than a claim to reproduce a task-specific vision or language system [6, 12, 13].

The recovery-oriented multi-expert router (RO-MER) is a stronger sample-dependent control motivated by recent failure-resilient expert routing [14]. Its candidate set contains one classifier from each physical group and one joint expert that maps all group representations, availability indicators and quality values through a 64-unit hidden layer. A separate 64-unit router produces probabilities $\alpha_{i1}, \ldots, \alpha_{iM}, \alpha_{iJ}$ over the available group experts and the joint expert. The prediction uses this unmodified $(M + 1)$-expert distribution. For losses that require physical-group weights, the differentiable effective allocation is

$$w_{im}^{\text{eff}} = \alpha_{im} + \alpha_{iJ} \frac{a_{im}}{\sum_k a_{ik}}. \tag{13}$$

The joint probability is therefore distributed uniformly only for the shared monotonicity and gate-regularization terms. This allocation is not detached, but it does not alter the routed prediction. RO-MER has 35,811 trainable parameters and receives the same faults, batches, seeds and checkpoint rule as RO-CAGF and RO-PDRF-Full.

The complete RO-MER objective is

$$\begin{aligned} \mathcal{L}_{\text{RO-MER}} = \mathcal{L}_{\text{task}} + 0.15\mathcal{L}_{\text{deg}} + 0.10\mathcal{L}_{\text{cons}} + 0.05\mathcal{L}_{\text{mono}} \\ + 0.30\mathcal{L}_{\text{rec}} + 0.05\mathcal{L}_{\text{Brier}} + 0.10\mathcal{L}_{\text{AUC}} + 0.005\mathcal{L}_{\text{router}}, \end{aligned} \tag{14}$$

where $\mathcal{L}_{\text{router}}$ is the mean clean-plus-degraded Kullback–Leibler divergence from $\mathbf{w}_i^{\text{eff}}$ to the uniform distribution over available physical groups. RO-CAGF uses the same coefficients and terms, with its native M-group softmax weights replacing Eq. (13). RO-PDRF-Full replaces $\mathcal{L}_{\text{router}}$ with the response-interior penalty because its bounded responses have a different geometry. Thus recovery supervision, proper scoring, pairwise ranking, paired views and task losses are matched, whereas only the architecture-specific regularizer differs. RO-MER tests whether the conclusion survives explicit cross-group interaction rather than merely widening the original gate.

We additionally adapt Quality-aware Multimodal Fusion (QMF) to the same four sensor groups while retaining its detached energy-confidence fusion and historical-loss ranking rule [18]. This QMF-PD control shares the present encoders, corruptions, optimizer, split and checkpoint criterion; it is a sensor-group adaptation rather than an exact reproduction of the original backbones or tasks. Implementation choices and sensitivity variants are specified in the Supplementary Information.

PDRF contains 19,740 trainable parameters on the chemical array, compared with 26,588 for CAGF and DWR, 35,811 for RO-MER, 26,182 for early fusion and 17,560 for uniform fusion. Ablations comprise paired degradation only (PD-ONLY), paired degradation plus normalized-weight monotonicity (PD-MONO), paired degradation plus response ranking (PD-RANK), the complete objective without $\mathcal{L}_{\text{score}}$ (NO-LSCORE), and $\mathcal{L}_{\text{score}}$ without $\mathcal{L}_{\text{rank}}$ (LSCORE-NO-LRANK). A PDRF-NOQ model is trained and tested with all quality values fixed to one.

3.5 Recovery-oriented extension

The mechanism audit showed that reducing an affected group's weight was often necessary but not sufficient for prediction recovery. We therefore define recovery-oriented PDRF (RO-PDRF), which retains the PDRF architecture and adds recovery supervision. First, the degraded prediction is distilled from the detached clean-view prediction. Write $\widehat{\mathbf{p}}_i = \text{stopgrad}(\mathbf{p}_i)$; then

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{V}_{\mathcal{B}}|} \sum_{i \in \mathcal{V}_{\mathcal{B}}} D_{\text{KL}}\left[\widehat{\mathbf{p}}_i \,\|\, \mathbf{p}'_i\right]. \tag{15}$$

As with the removal consistency term, $\mathcal{L}_{\text{rec}}$ is defined only for observations with at least two available groups, because removing the degraded group from a single-group observation leaves no valid recovery view; it is omitted when $\mathcal{V}_{\mathcal{B}}$ is empty. In the frozen chemical run, this eligibility rule excluded $0.386\% \pm 0.052\%$ of 5,933 training rows across seeds. The exclusion rate and counts are reported in the Supplementary Information. This label-free teacher targets recovery of the clean class distribution instead of assuming that downweighting alone repairs the decision. We denote the base PDRF objective plus $0.30\mathcal{L}_{\text{rec}}$ as RO-PDRF-Lite. Label-assisted, confidence-weighted, exponential-moving-average and agreement-gated teachers are sensitivity controls rather than selector inputs [58]. Their definitions, decay grid and conditional recovery/error-transfer results are reported in the Supplementary Information; no held-out label is used by the deployed selector.

Second, multiclass Brier losses on clean and degraded predictions provide proper-scoring supervision,

$$\begin{aligned} \mathcal{L}_{\text{Brier}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \|\mathbf{p}_i - \mathbf{e}_{y_i}\|_2^2 \\ + \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \|\mathbf{p}'_i - \mathbf{e}_{y_i}\|_2^2. \end{aligned} \tag{16}$$

The Brier term is the unweighted multiclass score averaged over observations; class weights are used for cross-entropy but are not inserted into the proper score. This preserves its standard probabilistic interpretation while classwise calibration is reported separately. Third, a weak log barrier discourages boundary occupancy without removing the original finite quartic penalty. With $\epsilon = 10^{-4}$, define $b_\epsilon(r) = -\log\{\max[\epsilon, 1-(r/B)^2]\}$, $\bar{b}_{im} = b_\epsilon(r_{im}) + b_\epsilon(r'_{im})$ and $Z_{\mathcal{B}} = \sum_{i \in \mathcal{B}} \sum_m a_{im}$. Its exact clean-plus-degraded implementation is

$$\mathcal{L}_{\text{interior}} = Z_{\mathcal{B}}^{-1} \sum_{i \in \mathcal{B}} \sum_m a_{im} \bar{b}_{im}. \tag{17}$$

Unavailable groups are excluded by a_{im}; the clamp is solely a numerical safeguard near the open boundary. Fourth, a smooth one-versus-rest pairwise ranking loss directly supervises degraded-view class ordering. For class c, let P_c and N_c denote positive and negative observations in a minibatch, let $\mathcal{C}_\mathcal{B} = \{c : |P_c| > 0, |N_c| > 0\}$ be the eligible class set, and let $\phi_{ijc} = \text{softplus}[-(\ell'_{ic} - \ell'_{jc})]$:

$$\begin{aligned}\mathcal{L}_{\text{AUC}} &= \frac{1}{|\mathcal{C}_\mathcal{B}|} \sum_{c \in \mathcal{C}_\mathcal{B}} A_c, \\ A_c &= \frac{\sum_{i \in P_c} \sum_{j \in N_c} \phi_{ijc}}{|P_c||N_c|}.\end{aligned} \tag{18}$$

Pairwise losses target class ordering directly rather than using cross-entropy as an indirect surrogate [59]. A class contributes only when the current minibatch contains at least one positive and one negative example; absent classes are skipped, and the pairwise term is omitted if no class is eligible. RO-PDRF-Full adds $0.30\mathcal{L}_{\text{rec}} + 0.05\mathcal{L}_{\text{Brier}} + 0.005\mathcal{L}_{\text{interior}} + 0.10\mathcal{L}_{\text{AUC}}$ to Eq. (12). The formal hierarchy is fixed throughout: RO-PDRF-Lite is the primary practical model and software default; RO-PDRF-Full is the mechanistic full model used for complete-objective architecture comparisons; and RO-PDRF-EMA is a calibration-oriented sensitivity based on Full. The clean-view teacher is the formal specification because it removes label-assisted teacher selection without reducing affected-subset discrimination in the teacher audit. The test set is not used to fit model or ensemble parameters.

To separate recovery supervision from architecture, we also train recovery-oriented CAGF (RO-CAGF). It receives the same clean-view distillation, Brier and degraded-view ranking losses as RO-PDRF-Full. Its unrestricted softmax gate uses a weak KL-to-uniform regularizer (weight 0.005), whereas RO-PDRF-Full uses the response-interior penalty. The matched data, fixed faults, seeds, minibatches and checkpoint rules isolate architecture subject to this necessary regularizer difference. Regularizer grids, early/uniform-fusion controls, modality-dropout self-distillation and confidence-thresholded distillation are reported in the Supplementary Information [5, 6].

3.6 Cross-fitted selective recovery

Unconditional recovery can repair a wrong base prediction but can also overturn a correct one. We therefore treat routing as hard-decision risk control. Here "Lite" means that the router selects between PDRF and the RO-PDRF-Lite recovery endpoint; it does not merely denote a smaller selector. Both endpoints require one ordinary forward pass and have the same 19,740-parameter architecture. The heavier PDRF/RO-PDRF-Full pair is retained only as a supplementary mechanistic and legacy-routing sensitivity.

Batch 7 is divided chronologically into two disjoint halves. The first half fits one scalar temperature for each endpoint by minimizing multiclass negative log-likelihood. The second half contains two calibration-only realizations of Gaussian, offset, drift and stuck-at corruption. Only rows on which the intervention reaches an available affected group enter selector development. Every routing feature is calculated from these temperature-scaled endpoint probabilities. The emitted vector is the selected endpoint's temperature-scaled vector; no post-routing recalibration is fitted.

Lite-CF uses six prespecified inputs: base confidence, recovery confidence, their difference, base normalized entropy, recovery normalized entropy and their difference. It receives neither the class label nor fault type at inference. Let z_{ij} denote standardized feature j. Feature means and standard deviations are fitted only on the informative training rows inside each cross-fitting fold, then refitted on all informative calibration rows for the final model. An L_2-regularized logistic model produces the *conditional preference score*

$$\pi_i = \sigma\left(\gamma_0 + \sum_{j=1}^{6} \gamma_j z_{ij}\right), \tag{19}$$

using calibration rows on which PDRF and RO-PDRF-Lite differ in correctness. The target equals one when recovery is correct and the base endpoint is wrong. This conditional score is not a calibrated reliability probability for the full stream. Lite-CF uses $C = 0.05$, the `lbfgs` solver, balanced class weights, tolerance 10^{-4}, 2,000 maximum iterations and random state 0.

Coefficient fitting is separated from threshold selection by five-fold stratified group cross-fitting. All realizations and mechanisms originating from the same calibration observation remain in one fold. Candidate thresholds are $0.50, 0.525, \ldots, 0.90$ and are evaluated only on pooled out-of-fold scores. Selection is lexicographic: minimize realized harms, maximize selected accuracy, maximize retained corrections and, if all three remain tied, choose the highest threshold. The formal Lite-CF fits had a mean of 1.5 tied thresholds (range 1–4). The final coefficients are refitted on all informative rows while the threshold remains frozen. A newly trained deployment model follows the same reserved chronological-calibration protocol. If both preference classes are not present in every grouped fold, the selector is declared unsupported and routing falls back to PDRF; test observations are never borrowed to complete a fit.

We probe the selected-sample and sparse-event risks in four ways. First, ten repeated grouped splits are run for each of 10 fitted endpoint pairs. Second, standardized coefficient distributions, sign frequencies and $C \in \{0.01, 0.05, 0.1, 0.5, 1, 5\}$ are audited. Third, calibration groups are subsampled to 25%, 50%, 75% and 100%. Fourth, an all-row multinomial model predicts three outcomes—base better, equivalent or recovery better—from the same six features. A depth-two tree with minimum leaf size 35 supplies a nonlinear, compute-matched learned baseline. These models use the same grouped cross-fitting and calibration budget as Lite-CF.

Legacy Safe-CF-12, staged-cascade, scalar-rule, random-selection and correctness-opportunity controls are defined and evaluated in the Supplementary Information. They are not part of the primary two-pass deployment chain.

The deployment audit reuses the same endpoints, temperatures, selector and test realization for every stream. It includes the strict fault-applied-and-available subset, a complete 40%-prevalence mixed stream and a no-imposed-fault stream with the same fixed 20% group-missingness process. Probability quality is reported separately for endpoint-class agreement and disagreement. The primary outcomes are negative-transfer prevention, recovery retention and

$$U = V_C N_C - C_H N_H - \lambda(P - 1), \tag{20}$$

where N_C and N_H are retained corrections and realized harms per 10,000 observations, V_C is correction value, C_H is harm cost, P is mean forward-pass count and λ is a transparent pass-cost proxy. We vary C_H/V_C over 1, 2, 5 and 10, the imposed-fault prevalence, and λ. Energy was not measured, so forward passes and counted linear-layer FLOPs are reported only as computational proxies.

3.7 Chemical-sensor dataset and fixed faults

The UCI Gas Sensor Array Drift Dataset contains 13,910 observations from 16 metal-oxide sensors, six gas classes and 10 acquisition batches over 36 months [60, 61]. Each sensor contributes eight extracted response features. The principal grouping uses four consecutive sensors per group. All models receive the same grouping. Sensitivity analyses additionally use interleaved and five fixed random partitions; no partition is called a physical modality.

Batches 1–6 train the models. Their feature means and standard deviations are the only quantities used for standardization. The same transform is applied to batches 7–10, after which each feature is clipped to $[-8, 8]$ before sensor grouping. Thus every reported "training range" is the empirical minimum–maximum after training-only standardization and clipping; it is not a physical sensor limit. The first 60% of Batch 7 selects checkpoints, and the remaining 40% is reserved for calibration; the routing experiment further divides that calibration partition into temperature and selector halves. Batches 8–10 are never used to fit preprocessing, endpoints, temperatures, selectors or thresholds.

Controlled faults are imposed after this training-only preprocessing in grouped standardized space. The formal test assigns 40% of observations a scale-3 intervention on group 1 and fixes $q = 1$. Fault assignment and 20% group missingness are generated independently to isolate their factorial effects. This is a stress design, not a claim that all missingness–corruption combinations occur naturally. Of 4,364 test observations, 1,765 are assigned the intervention; group 1 remains available and receives it in 1,436, whereas it is masked in 329. The strict recovery estimand is the former *fault-applied-and-available* subset. Fault indices, missingness masks and intervention values are fixed across methods and optimization seeds.

The complete test set contains 294, 470 and 3,600 observations in batches 8, 9 and 10. Its six class counts are 691, 685, 740, 708, 821 and 719. The corresponding strict-subset batch counts are 91, 165 and 1,180, and its Gaussian class counts are 230, 220, 230, 253, 266 and 237. Complete mechanism-by-batch class tables are supplied in the Supplementary Information.

Checkpoint selection and temperature calibration use separate fault-free views with independently generated 10% group missingness; no controlled Gaussian, offset, drift or stuck-at intervention is added to either subset. The checkpoint criterion is unweighted cross-entropy on the complete selection subset. For the cross-fitted deployment run, temperature is fitted on the first chronological half of the calibration partition, not an affected subset, by minimizing unweighted multiclass negative log-likelihood. The resulting scalar is specific to each fitted seed and method and is applied unchanged to every deployment stream. Controlled-fault and no-fault deployment streams use the same fixed 20% group-missingness process so that fault prevalence is the changing factor. Earlier natural-view analyses without imposed test missingness remain supplementary sensitivity evidence.

The expanded fault study crosses four prespecified mechanisms with realizations 70001–70008 and fitted seeds 101–105. Model fitting always uses the Gaussian paired degradation defined above; offset, drift and stuck-at are test-time interventions only. Gaussian faults add independent standardized noise. Offset faults add a featurewise signed displacement. Drift multiplies that displacement by a monotone 0.25–1.00 trajectory over held-out acquisition order. Stuck-at faults replace the affected group's standardized features by a fixed signed vector. Within each realization, all mechanisms share affected rows and missingness masks, while intervention values remain fixed across methods. These are controlled repeats on one instrument.

Table 2 translates the formal scale into empirical training-distribution displacement. At scale 3, median absolute shifts were 1.89–3.10 training standard deviations and median percentile shifts were 0.36–0.56. Only 9.4–45.7% of corrupted values remained inside the training 1st–99th percentile interval. The formal setting is therefore a severe standardized stress test, not a typical natural-failure model. Scale-1 and scale-5 summaries and representative inverse-transformed feature examples are reported in the Supplementary Information.

Exploratory controls vary diagnostic-quality metadata, leave-one-group-out training, response bounds, weighting slope, ranking weight, fault group, severity and prevalence. They share observations and fitted models, are not treated as independent replications and do not replace the prespecified main setting. Full specifications and results are reported in the Supplementary Information.

Table 2: Fault-plausibility audit in the strict subset. Fractions are calculated featurewise against batches 1–6. “Training range” denotes the observed training minimum–maximum, not a physical sensor limit.

Mechanism (scale 3)	Median $\lvert\Delta z\rvert$	Median percentile shift	Within training 1–99%	Within training range
Gaussian	2.015	0.360	0.379	0.609
Offset	3.000	0.538	0.332	0.632
Drift	1.888	0.440	0.457	0.697
Stuck-at	3.105	0.557	0.094	0.656

3.8 Independent hydraulic single-rig sensitivity analysis

The UCI Condition Monitoring of Hydraulic Systems dataset contains 2,205 raw 60-s cycles from an experimental hydraulic rig and native condition labels for cooler, valve, pump and accumulator states (DOI: 10.24432/C5CW21) [22, 62]. We retain 1,449 cycles marked as stable. Fourteen physical sensors form five groups by measured quantity: six pressure sensors, two flow sensors, four temperature sensors, motor power and vibration. This grouping is defined before model fitting and follows the measurement system rather than a random partition.

Each sensor is summarized by distributional and trend statistics. Unequal group lengths are aligned to 48 inputs; a zero-padded native-order representation audits possible smoothing. The main split preserves acquisition order within every native condition class, assigns contiguous 60%/15%/10%/15% blocks and removes five cycles at each boundary. A global chronological split would omit some condition classes from training. After guard-band removal, the test counts are 72/72/74 for the three cooler states, 54/54/54/56 for the four valve states, 74/72/72 for the three pump states and 56/54/54/54 for the four accumulator states. Recovery-objective-matched RO-PDRF-Full and RO-CAGF use five matched seeds; controlled pressure- and vibration-group faults affect the same 40% of test cycles. Component-condition labels are native to the rig, whereas sensor faults remain interventions. Representation and random-split sensitivities are supplementary [23–25].

3.9 Real-operational AHU direct-diagnosis boundary analysis

We additionally use the open real-operational AHU dataset from an auditorium, a hospital and a large office (Figshare DOI: 10.6084/m9.figshare.27147678.v3) [26, 27]. The binary task uses native, expert-annotated temperature-sensor-fault labels and no artificial test corruption. Seven measurements form five physical groups; each building uses a chronological 60%/15%/10%/15% split and five matched seeds. The untouched test periods contain 228/9,354 fault records in the auditorium (2.44%), 1,495/9,362 in the hospital (15.97%) and 346/19,115 in the office (1.81%). Return-air/supply-air subtype counts are 179/49, 1,487/8 and 293/53, respectively. Because the label is the sensor fault itself and no independent downstream state is available, this is a negative task-boundary audit, not downstream recovery validation. Detailed shift, adaptation, risk–coverage and conformal analyses are reported in the Supplementary Information [41–44].

3.10 Training, metrics and statistical analysis

AdamW uses learning rate 2×10^{-3}, weight decay 10^{-4}, batch size 256 and gradient clipping at 5. Chemical-array training uses at most 60 epochs, hydraulic training uses at most 80 and AHU training uses at most 12. Checkpoints minimize unweighted selection cross-entropy and stop after nine epochs without an improvement greater than 10^{-4}. RO-PDRF-Lite, RO-PDRF-Full, RO-

PDRF-EMA and all teacher controls use this identical criterion; no test metric selects a checkpoint or variant. Class weights are $n/(Cn_c)$. No dropout or batch normalization is used.

The primary deployment outcomes are negative-transfer prevention, recovery retention and the utility in Eq. (20). Macro-AUROC, macro-AUPRC, NLL, multiclass Brier score and calibration are secondary. Equal-width ECE partitions maximum class probability into 15 intervals on $[0, 1]$. Equal-mass ECE splits the sorted maximum probabilities into 15 groups. Classwise ECE applies 15 equal-width bins to each one-versus-rest probability and averages the six values. Classwise AUROC, AUPRC and recall are archived. Temperature scaling is fitted separately for each endpoint and optimization seed on the disjoint first half of Batch 7.

The deployment analyses are frozen as CEE-CF10-R2 for the Full endpoint and CEE-CF10-R2-LITE for the primary Lite endpoint. Within each seed, one endpoint pair is reused across all four faults and selector comparisons. Test realization 70001 is fixed. Means over the strict set give equal weight to the four mechanism–seed cells. Clean/no-fault summaries contain one value per fitted pair. Mechanism-specific effects use paired Student-t intervals across the 10 fitted model pairs while holding the mechanism fixed. Batch-aware sensitivity holds both mechanism and acquisition batch fixed. These intervals quantify fitting variation on the present data; they are not device- or fault-population intervals.

The previous four-mechanism crossed bootstrap is retained only as a labelled sensitivity in the Supplementary Information. It is not used for new headline inference because four top-level mechanism clusters do not support a reliable mechanism-population bootstrap. Exact Wilcoxon tests across optimization seeds are likewise reported only as fitting-stability diagnostics. Hydraulic summaries use a task–fault–seed hierarchy; AHU results remain building-specific. The strict estimand and selector analyses followed reviewer-driven audit and are not described as preregistered confirmatory analyses.

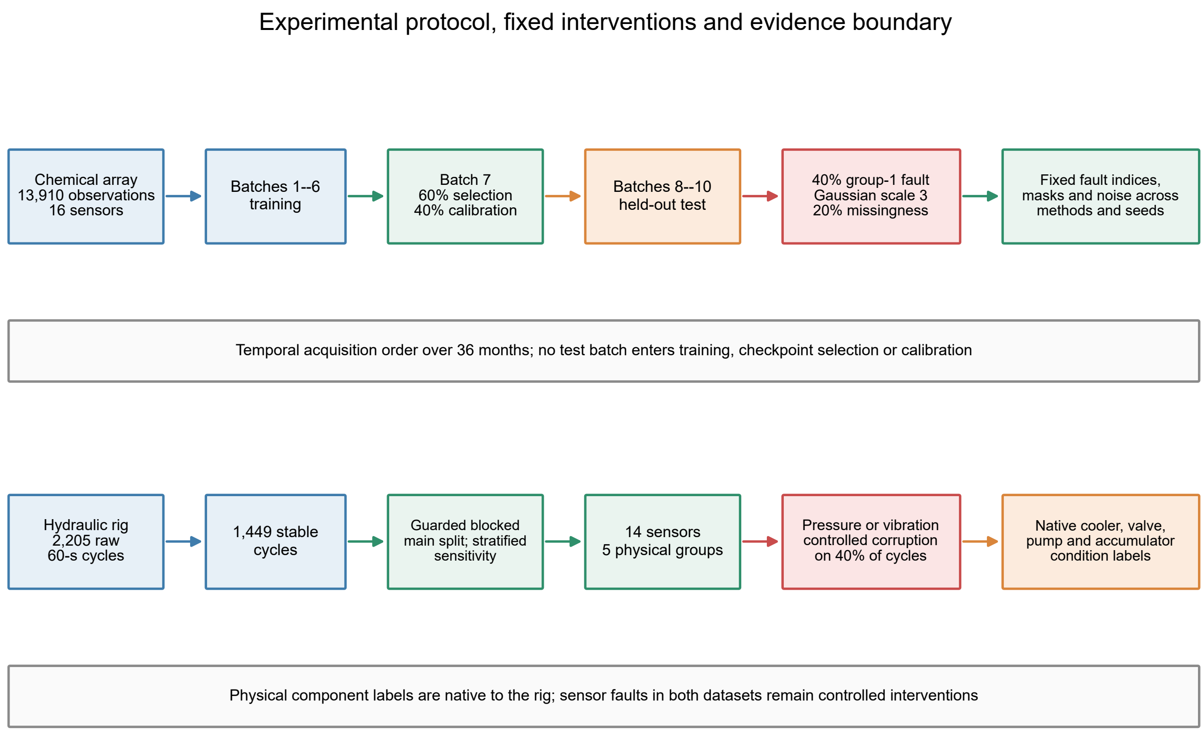

Figure 2: **Integrated experimental protocol and controlled fault design.** The upper lane follows the chemical-array acquisition order from batches 1–6 training through Batch 7 checkpoint selection/calibration to held-out batches 8–10; fault indices, masks and intervention values are fixed across methods and seeds. The lower lane follows the independent hydraulic rig from stable 60-s cycles to the guarded within-condition blocked main analysis; the earlier stratified split is retained only as a sensitivity analysis. Component-condition labels are native to the rig, whereas chemical, pressure and vibration sensor corruptions remain controlled interventions.

4 Results

4.1 Primary two-pass routing controlled hard-decision risk

The strict subset contains 1,436 observations per mechanism on which the assigned intervention reached an available group. Across the 40 mechanism–seed cells, PDRF achieved mean accuracy 0.4869 and macro-AUROC 0.8115. Unconditional RO-PDRF-Lite increased these values to 0.5002 and 0.8175. Lite-CF emitted the recovery output for 52.2% of observations, prevented 94.7% of recovery-induced negative transfers and retained 9.7% of the corrections available from unconditional recovery. Its accuracy and macro-AUROC were 0.4885 and 0.8151 (Table 3). Thus the router achieved its hard-decision safety target but did not preserve most of the recovery endpoint's accuracy gain.

The routed probability vector improved on PDRF for NLL, Brier score, ECE and macro-AUPRC, but remained below unconditional RO-PDRF-Lite for every strict-set probability metric. On the complete 40%-prevalence stream, Lite-CF/PDRF/RO-PDRF-Lite macro-AUROC values were 0.8441/0.8426/0.8451, and NLL values were 2.1386/2.2131/2.1135. On the no-imposed-fault stream, their macro-AUROCs were 0.8629/0.8631/0.8623 and NLL values were

Table 3: **Strict-subset endpoint and routed-output quality.** Values are equal-weight means over four fixed mechanisms and 10 fitted model pairs. Each selector had 89.7 informative endpoint-correctness disagreements on average; thresholds from 100 repeated grouped fits ranged from 0.525 to 0.90. Probabilities are temperature-scaled before routing; no probability mixture or post-routing recalibration is applied. ECE uses 15 equal-width bins.

Method	Accuracy ↑	AUROC ↑	AUPRC ↑	NLL ↓	Brier ↓	ECE ↓
PDRF	0.4869	0.8115	0.5534	2.8239	0.8286	0.3492
RO-PDRF-Lite	**0.5002**	**0.8175**	**0.5643**	**2.6023**	**0.7988**	**0.3302**
Lite-CF	0.4885	0.8151	0.5598	2.6775	0.8129	0.3408

1.8636/1.9002/1.8626. Lite-CF prevented 94.2% of no-fault negative transfers but retained only 1.7% of no-fault corrections, giving -5.3 net correct decisions per 10,000. A low residual-harm rate—the observation-level form of a harmful switch—therefore does not imply that the routed probability vector is uniformly better.

Mechanism-fixed paired intervals replace the four-cluster bootstrap as the main inferential display (Table 4). Lite-CF gains over PDRF were clearest for Gaussian and stuck-at corruption at the tested severity. The offset interval was effectively centred near the null and crossed zero; the drift interval also crossed zero. Gaussian Lite-CF gains were positive within batches 8, 9 and 10 (+0.0062, +0.0077 and +0.0061), whereas most offset and drift batch-fixed intervals crossed zero. The result is mechanism-dependent and cannot be interpreted as a uniform corruption-family or fault-population effect.

Table 4: **Mechanism-fixed macro-AUROC differences.** Brackets are paired 95% Student-t intervals across 10 fitted model pairs while holding the mechanism and test observations fixed. Mechanisms are not resampled as a population.

Fixed mechanism	RO-PDRF-Lite − PDRF	Lite-CF − PDRF
Gaussian	$+0.0124$ $[+0.0102, +0.0146]$	$+0.0064$ $[+0.0034, +0.0093]$
Offset	$+0.0034$ $[+0.0007, +0.0060]$	$+0.0028$ $[-0.00002, +0.0057]$
Drift	$+0.0020$ $[-0.0005, +0.0044]$	$+0.0018$ $[-0.0002, +0.0037]$
Stuck-at	$+0.0064$ $[+0.0033, +0.0095]$	$+0.0034$ $[+0.0005, +0.0063]$

At equal correction value and harm cost and without a pass penalty, Lite-CF retained 24.20 corrections and realized 8.36 harms per 10,000 strict observations, yielding net utility +15.84. The net values at harm-to-correction ratios 2, 5 and 10 were +7.49, -17.58 and -59.37. The approximate 2.9 break-even ratio is a descriptive, dataset-specific boundary for this endpoint pair, event definition, test distribution and fault prevalence. Table 5 further crosses complete-stream fault prevalence with the pass-cost proxy. A deployment preference is defined only after the application supplies correction value V_C, harm cost C_H and compute penalty λ; energy was not measured.

4.2 Sparse calibration limited selector stability

The formal Lite-CF calibration sets contained a mean of 1,861 rows, but only 89.7 were correctness disagreements: 36.1 favoured PDRF and 53.6 favoured RO-PDRF-Lite. Frozen thresholds averaged 0.710 and ranged from 0.55 to 0.90. Repeating the grouped cross-fitting 10 times for each end-

Table 5: **Fault-prevalence and pass-cost sensitivity.** Values are means across the complete deployment streams at equal correction value and harm cost ($V_C = C_H = 1$). λ is the cost of one extra forward pass in correction-equivalents per 10,000 observations. These values are descriptive for the present endpoints and controlled test distribution.

Fault prevalence	Corrections/10,000	Harms/10,000	$U(\lambda = 0)$	$U(\lambda = 0.5)$	$U(\lambda = 2)$
0%	1.37	6.65	-5.27	-5.77	-7.27
10%	3.27	6.93	-3.67	-4.17	-5.67
40%	8.88	7.33	1.55	1.05	-0.45
70%	15.98	8.36	7.62	7.12	5.62

point pair produced mean out-of-fold AUROC 0.720 (standard deviation 0.138; range 0.327–0.954), AUPRC 0.809 and Brier score 0.208. Thresholds ranged from 0.525 to 0.90. Numerical convergence therefore did not imply stable conditional discrimination.

Coefficient signs were most stable for Lite confidence (94%), base confidence (89%) and confidence change (83%). Dominant-sign fractions for the entropy terms were 61–67%. Strong shrinkage traded away conditional discrimination: mean out-of-fold AUROC increased from 0.690 at $C = 0.01$ to 0.717 at $C = 0.05$, 0.800 at $C = 1$ and 0.811 at $C = 5$. The prespecified $C = 0.05$ is therefore a conservative estimation choice, not the value that maximizes calibration preference AUROC.

Calibration-size sensitivity showed the same boundary. Using 25%, 50%, 75% and 100% of calibration groups produced pooled mean prevention of 88.7%, 91.6%, 92.4% and 93.9%, respectively. Retention changed from 12.1% to 10.0%, while macro-AUROC remained near 0.81. These are repeated-subsampling summaries and differ from the equal-weight 40-cell headline estimand.

The all-row three-outcome model and shallow tree did not dominate Lite-CF. Legacy Full-endpoint transport, staged-cascade, scalar-rule and correctness-opportunity comparisons likewise exposed unresolved trade-offs among safety, recovery, ranking and compute. Their complete estimands and tables are moved to the Supplementary Information so that the primary chain remains the PDRF–RO-PDRF-Lite–Lite-CF comparison.

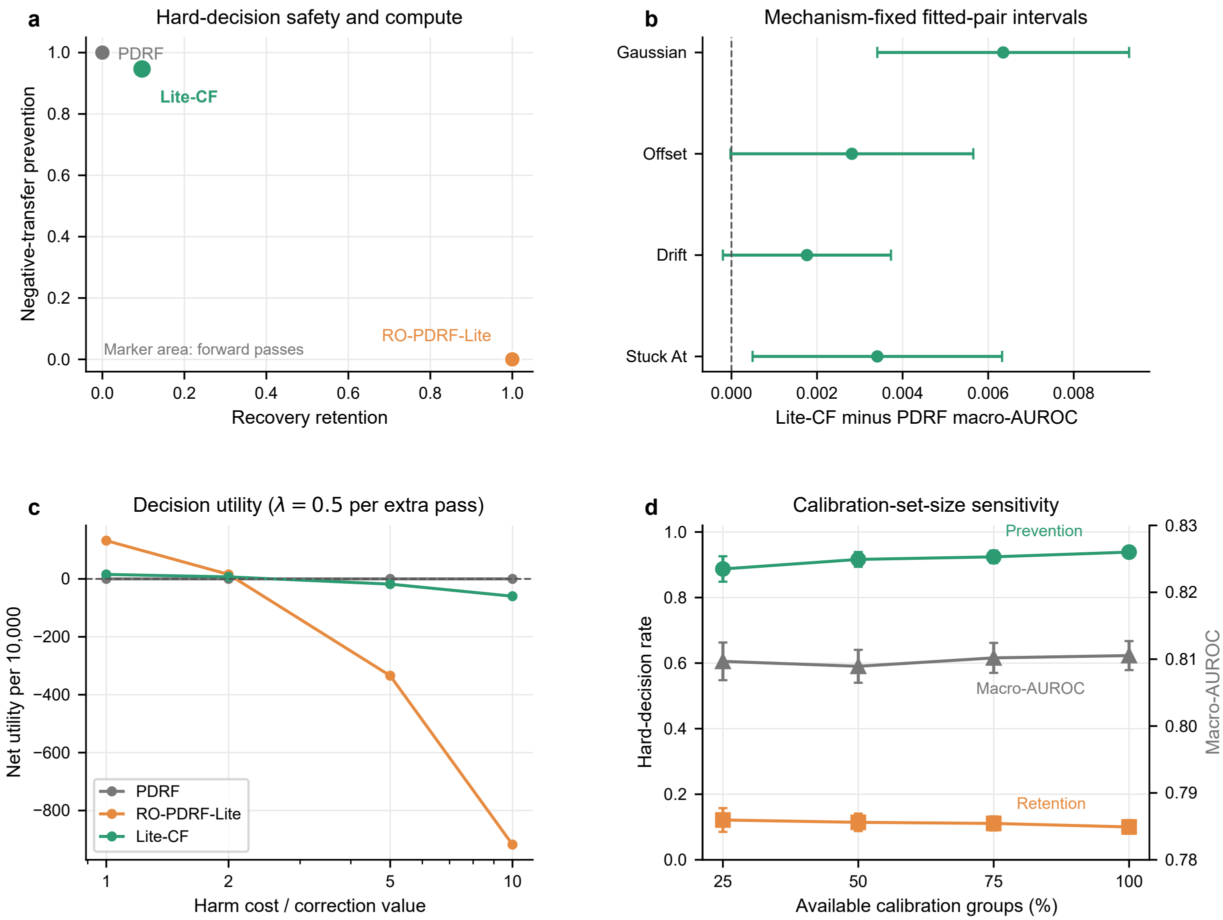

Figure 3: **Primary safety, mechanism and calibration evidence.** **a**, negative-transfer prevention versus recovery retention for PDRF, unconditional RO-PDRF-Lite and Lite-CF on the strict subset; marker area scales with mean forward passes. **b**, Lite-CF minus PDRF macro-AUROC with each mechanism held fixed; bars are paired 95% Student-t intervals across 10 fitted model pairs. **c**, net utility per 10,000 strict observations with an exact pass penalty of $\lambda = 0.5$ correction-equivalents per 10,000 for each extra forward pass; this is a computational proxy, not measured energy. **d**, repeated calibration-group subsampling; bars show 95% intervals over seed–repeat summaries.

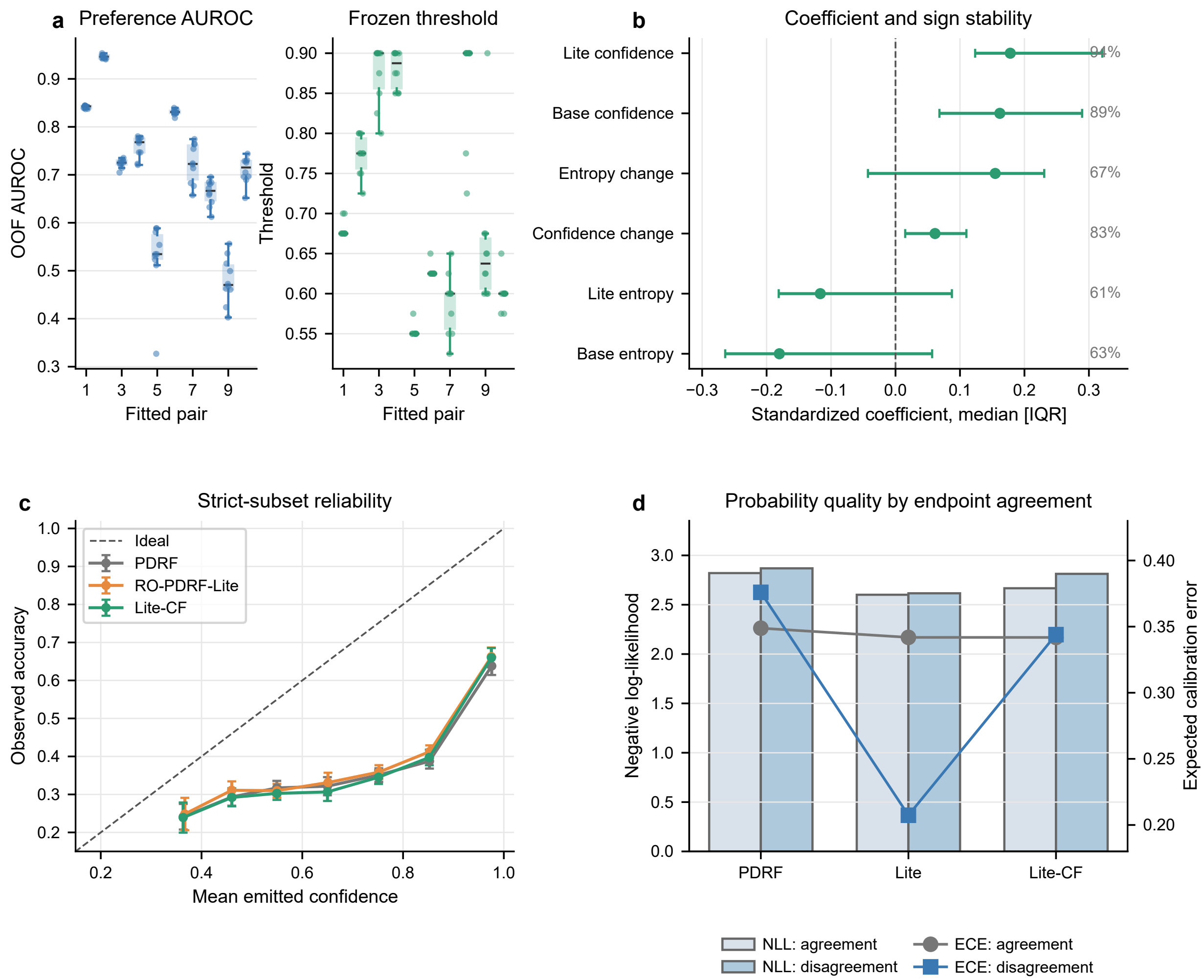

Figure 4: **Selector stability and final probability quality.** **a**, separate seed-wise distributions of out-of-fold preference AUROC and selected threshold across 100 repeated grouped fits; boxes and points prevent the two quantities from being visually conflated. **b**, median standardized Lite-CF coefficient with interquartile range; right-side percentages are dominant-sign frequencies. **c**, reliability curves for strict-subset outputs. **d**, negative log-likelihood bars and 15-bin expected calibration error lines after separating endpoint-class agreement from disagreement. Each endpoint vector is temperature-scaled before routing; Lite-CF emits one selected endpoint vector, not a probability mixture, and no post-routing recalibration is fitted.

4.3 Hydraulic single-rig sensitivity boundary

The guarded hydraulic analysis used four native component-condition tasks and controlled pressure- and vibration-group faults (Table 6). RO-PDRF-Full exceeded recovery-objective-matched RO-CAGF in six of eight task–fault conditions. The mean difference was +0.015, but the task–fault–seed hierarchical interval (−0.006 to +0.036) included zero. Performance ranged from near ceiling for cooler (RO-PDRF-Full macro-AUROC 0.996) to near chance for pump and accumulator (0.528 and 0.487), limiting architecture interpretation on the latter tasks.

The earlier random-stratified split yielded a larger mean difference (+0.044), indicating optimistic adjacent-cycle allocation. Guard bands reduced direct adjacency but did not create independent devices. No-interpolation and random-split results are therefore supplementary sensitivity evidence; the main result supports single-rig platform-internal robustness, not a general architecture ranking.

Table 6: **Hydraulic single-rig sensitivity analysis.** Guarded-block affected-subset macro-AUROC values are means ± standard deviations across five matched seeds. The eight task–fault rows are conditions from one rig, not independent devices; the earlier random-stratified results are supplementary sensitivity evidence.

Task	Fault	RO-CAGF	RO-PDRF-Full	Difference
Cooler	Pressure	0.942 ± 0.023	0.992 ± 0.004	+0.050
Cooler	Vibration	0.947 ± 0.029	1.000 ± 0.000	+0.053
Valve	Pressure	0.730 ± 0.026	0.762 ± 0.023	+0.032
Valve	Vibration	0.739 ± 0.029	0.749 ± 0.032	+0.010
Pump	Pressure	0.569 ± 0.035	0.536 ± 0.018	-0.033
Pump	Vibration	0.533 ± 0.061	0.521 ± 0.022	-0.012
Accumulator	Pressure	0.478 ± 0.022	0.491 ± 0.023	+0.013
Accumulator	Vibration	0.474 ± 0.047	0.483 ± 0.044	+0.009

4.4 AHU direct-diagnosis boundary

The chronological air-handling-unit tests contained 37,831 untouched future-time records, including 2,069 expert-annotated temperature-sensor faults. Because the target is the fault itself and no independent downstream state is available, this experiment cannot test whether suppressing that sensor recovers another prediction. Early fusion exceeded bounded recovery in all 15 building–seed pairs (mean AUROC difference −0.0341 for bounded recovery, $P < 0.001$), and leave-one-building-out bounded recovery collapsed to 0.298 ± 0.005 in the hospital. The operational data therefore establish a direct-diagnosis and transfer boundary, not downstream recovery validation; complete subtype, shift, adaptation and risk analyses remain in the Supplementary Information.

4.5 Routed probability quality depended on endpoint agreement

Final routed probabilities were evaluated directly rather than inferred from the residual-harm rate. On strict rows where PDRF and RO-PDRF-Lite predicted the same class, Lite-CF NLL/ECE were 2.667/0.342, between PDRF (2.821/0.349) and RO-PDRF-Lite (2.601/0.342). On endpoint-class disagreements, Lite-CF NLL/ECE were 2.813/0.344; unconditional Lite was better at 2.616/0.207, whereas PDRF was worse at 2.869/0.376. Lite-CF therefore reduced some probability loss without preserving the recovery endpoint's calibration advantage on disagreement cases (Fig. 4c,d). Equal-mass ECE, classwise ECE, classwise AUROC/AUPRC/recall and seed-level dispersion are reported in the Supplementary Information. No tested endpoint or router dominated hard-decision safety, ranking and calibration.

4.6 Lite-CF reduced, but did not remove, deployment cost

PDRF and RO-PDRF-Lite each require one forward pass, 19,740 stored parameters and 38,528 counted linear-layer FLOPs per four-group observation. Lite-CF stores both endpoints plus six standardized coefficients and an intercept. It requires two passes, 39,487 stored parameters and 77,056 endpoint FLOPs; the logistic operation is negligible relative to the endpoints. On an Intel Core i5-13500H with PyTorch 2.12.1+cpu and 12 threads, three warm-ups followed by seven timed calls gave mean seed-level median batch-one latencies of 0.734 ± 0.209 ms for PDRF, 0.748 ± 0.215 ms for RO-PDRF-Lite and 2.963 ± 0.569 ms for Lite-CF. Corresponding complete-batch (4,364 observations) latencies were 4.56 ± 0.59, 4.48 ± 0.51 and 12.50 ± 1.72 ms. These direct CPU values exclude loading and describe one workstation, not embedded or real-time performance. Energy, memory bandwidth and long-duration reliability were not measured. Lite-CF is therefore described only as two-pass and lower-compute than the six-pass legacy Safe-CF-12 sensitivity.

5 Discussion

The strict estimand and frozen-run discipline narrow the central interpretation. Clean-to-faulted distillation produced a useful recovery endpoint, but unconditional gains varied by mechanism. The contribution is therefore not a universal fusion architecture. It is a risk-control protocol that distinguishes fault assignment from an intervention that actually reaches an available group, measures both correction and error transfer, and freezes a cross-fitted routing threshold before test evaluation.

Cross-fitting addresses direct threshold leakage, not sparse-event estimation. Lite-CF had only 89.7 informative disagreements per endpoint pair on average. Repeated grouped fits showed wide preference-AUROC and threshold ranges, and several entropy coefficients changed sign. The six-feature, strongly shrunk model reduces dimensionality and compute, but it does not make the conditional score externally calibrated. Calibration is model- and deployment-specific. A site with too few examples from either preference class must therefore use PDRF while prospectively collecting new calibration evidence; a threshold from the present instrument should not be transferred unchanged.

The safety–recovery trade-off remains substantial. Lite-CF prevented 94.7% of negative transfers but retained only 9.7% of corrections, making it a conservative safety layer that sacrifices most recovery opportunities. Equal harm and correction values produced a positive strict-set result, whereas a harm cost above approximately 2.9 correction values reversed that dataset-specific preference. Complete-stream utility was negative at 0% and 10% controlled-fault prevalence and became positive at 40% and 70% under the same equal-cost model; a nonzero pass penalty shifted every value downward. Hard-decision safety, ranking, calibration and application-defined utility must therefore be reported separately. A low residual-harm rate alone is insufficient for deployment.

The two-pass design reduces the forward-pass count relative to the earlier six-pass endpoint hierarchy. It avoids leave-one-group-out views and uses the software-default RO-PDRF-Lite endpoint. Even so, it stores and evaluates two models. Direct workstation timing confirmed additional latency, especially at batch size one. These values do not establish embedded latency, memory-bandwidth adequacy, energy efficiency or real-time readiness; the utility analysis therefore retains transparent pass and FLOP proxies.

Mechanism-fixed and batch-fixed intervals replace the four-mechanism population bootstrap for headline inference. Gaussian gains were consistent across held-out batches. Offset and drift intervals often crossed zero. All intervals still reuse one chemical instrument and the same held-out observations; they quantify fitting variation, not independent devices. The formal scale-3 interventions are also severe standardized stress tests: most corrupted values fall outside the training 1st–99th percentile interval. This improves auditability but further limits natural-failure interpretation.

Transfer tests support specified controlled-corruption robustness, not natural hardware-failure or device transfer. The hydraulic rig supplies native downstream condition labels, yet the guarded architecture interval crosses zero and several tasks are near chance. It is single-rig sensitivity evidence. The AHU target is the sensor fault itself, so direct diagnosis rewards preserving an anomalous signature that recovery may suppress. A future field-recovery study should enroll multiple independently operated devices, adjudicate sensor failures from maintenance records and engineering review, and pair them with downstream control, energy, comfort or component labels that do not encode the sensor fault itself. Endpoint fitting and site-specific router calibration should be prospective, the base fallback should be predeclared for unsupported preference classes, and V_C, C_H and compute cost should be fixed before outcome analysis.

Additional boundaries remain. Response saturation prevents interval-scale device-health interpretation. Wrong clean teachers can propagate errors. No endpoint dominates probability quality and selective risk. Misleading quality metadata create another failure route. The evidence comes from controlled feature corruptions, one chemical instrument, one hydraulic rig and expert/rule-based AHU labels. Freezing the runs does not remove researcher-level adaptive-development risk. Prospective multi-device evaluation is required before field-recovery claims are warranted.

6 Conclusion

Cross-fitted selective recovery converted a useful but fallible recovery endpoint into an explicit deployment trade-off. On the strict controlled-corruption estimand, two-pass Lite-CF prevented 94.7% of negative transfers while retaining 9.7% of corrections; it is therefore a conservative safety layer, not a generally improved classifier. Utility was positive only over a bounded, dataset- and cost-model-specific region, and no-fault net correct change was slightly negative. Repeated grouped calibration exposed substantial threshold and coefficient instability, while mechanism-fixed and batch-fixed analyses showed that gains were not uniform across corruption types. Calibration must be repeated for each fitted deployment, and PDRF is the required fallback when both preference classes are unsupported. The selected endpoint probability vector was temperature-scaled but was not mixed or recalibrated after routing, and it was not uniformly better than both endpoints. The supported contribution is a reproducible hard-decision risk-control protocol for controlled available-group corruption, not evidence for calibrated device health, maintenance-confirmed natural failures or device-independent field recovery.

Data and code availability

The Gas Sensor Array Drift Dataset (DOI: `https://doi.org/10.24432/C5RP6W`) and Condition Monitoring of Hydraulic Systems dataset (DOI: `https://doi.org/10.24432/C5CW21`) are available from the UCI Machine Learning Repository under CC BY 4.0. The operational AHU data are available from Figshare under CC BY 4.0 (DOI: `https://doi.org/10.6084/m9.figshare.27147678.v3`). Code, fixed seeds, environment information, source data for CEE-CF10-R2 and figure/table scripts are publicly available in the `cee_revision` directory at `https://github.com/Corey9999/recovery-supervision-calibrated-fallback`. The frozen computational snapshot is commit `e200607e648dbbdd230b881f150de08a6da3e525`, archived under release tag `v1.1.0-cee-q1`. A permanent software DOI has not been assigned.

Ethics statement

The study used public engineering datasets and controlled computer-generated sensor corruptions. It involved no human participants, personal data, animals or biological specimens; institutional ethics review was not applicable.

Author contributions

Riyang Luo: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing–original draft, Writing–review and editing, and Project administration.

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Declaration of interests

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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